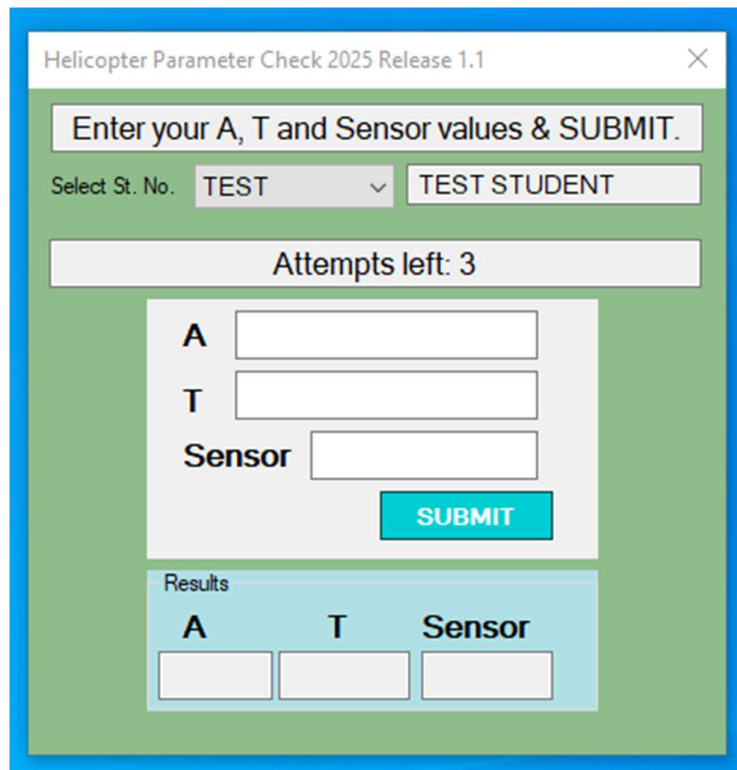


A brief walkthrough for checking your model

This doc will guide you on how to check the unknown parameters in your transfer function. This way, your model of the system will not be so wrong that you are unable to build a suitable controller/compensator that meets all the required specifications in both Lab 2 and 3.

Kitticopter Parameter Check software

When the application is opened and connected (this should take a while when opened for the first time), you should be met with the following screen.



The screenshot shows a software window titled "Helicopter Parameter Check 2025 Release 1.1". Inside the window, there is a green background with a white border. At the top, a white box contains the text "Enter your A, T and Sensor values & SUBMIT." Below this, there is a "Select St. No." label followed by a dropdown menu showing "TEST" and a "TEST STUDENT" button. A white box in the center displays "Attempts left: 3". Below this, there is a white box containing three input fields labeled "A", "T", and "Sensor", and a blue "SUBMIT" button. At the bottom, there is a "Results" section with three columns labeled "A", "T", and "Sensor", each with a corresponding input field.

The transfer function between the velocity output and the rotor thrust input should be first-order, meaning there should only be two unknowns in the open-loop transfer function, namely, the gain (A) and the time constant (T), as seen in the figure above. The final unknown is the sensor constant (k), which converts the Kitticopter's height in meters to a voltage in volts that is output by the DAC.

To check the accuracy of the parameters in your transfer function, you should first select your student no from the drop-down menu, then type your parameter values into their respective boxes and click 'submit' as shown below. This will give you an accuracy rating in % of how close your value was to the true value. The sensor value is quite trivial to obtain, so it should be quite accurate (>90%). As for the gain and time constant, you should aim for an accuracy of greater than 60%. However, it is important that you perform part 1 of the lab properly and try to get as accurate a model as possible the first time, as you only have **3 attempts** to check your transfer function.

Your results were recorded.

Select St. No.

TEST



TEST STUDENT

Attempts left: 2

A 0.7

T 0.9

Sensor 1.1

SUBMIT

Results

A

36 %

T

45 %

Sensor

45 %